

Optimal Measure Quantization for Machine Learning

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In this talk, I present a study on optimal measure quantization, which approximates continuous distributions by discrete measures supported on a finite number of points. This technique plays an important role in statistics and machine learning by reducing the complexity of optimization problems in high-dimensional spaces. The main goal is to analyze the relationship between joint quantization of several measures and the Wasserstein barycenter, exploring when both coincide and converge. The work combines theoretical and experimental perspectives, addressing the mathematics of quantization and optimal transport and discussing potential applications for future research.

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